

SECTION B [Fill in questions] - 14 marks

Write only your final, fully simplified answer in the block provided. Only this will be marked.

QUESTION 11 [2 marks]

Write the function $f(x) = |1 - 2x^2|$ as a piecewise defined function.

$f(x) =$	$1 - 2x^2$	if $x \in$ $[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]$
	$-(1 - 2x^2)$	if $x \in (-\infty, -\frac{1}{\sqrt{2}}) \cup (\frac{1}{\sqrt{2}}, \infty)$

QUESTION 12 [1 mark]

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{x} = \frac{2}{\pi}$$

QUESTION 13 [1 mark]

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} = \frac{1}{6}$$

QUESTION 14 [1 mark]

If $g(x) = 2 \ln(x) + 3x + 7$, then

$$g^{-1}(10) = 1$$

$$\begin{aligned} \text{let } g^{-1}(10) &= y \\ \Rightarrow g(y) &= 10 \end{aligned}$$

QUESTION 15 [2 marks]

Solve the inequality

$$\frac{|x - 3|}{x + 5} > 0$$

and write your solution using interval notation.

$$(-5, 3) \cup (3, \infty)$$

QUESTION 16 [2 marks]

Write as one expression:

$$\ln(x+5) + \frac{1}{2}(\ln x - \ln(x^2 + 6x + 5)^2) =$$

$$\ln\left(\frac{\sqrt{x}}{x+1}\right)$$

QUESTION 17 [2 marks]Complete the following, note that $\sin^{-1}(x) = \arcsin x$:

(a) $\sin^{-1}(x) = y \iff$

(b) $\sin^{-1}(\sin x) =$

for $x \in$

} See p.62 of
textbook.

QUESTION 18 [1 mark]

Write down the definition of an odd function.

See p. 16
(NB to mention that the equation must hold for all x in the domain of f .)

QUESTION 19 [2 marks]

Consider the piecewise defined function: $f(x) = \begin{cases} \cos x & \text{if } x \in [-\pi, 0), \\ \sin x + 3x & \text{if } x \in [0, \pi]. \end{cases}$ $\pi/2 \in [0, \pi]$

Determine the following limits, if they exist.

$$\lim_{x \rightarrow \frac{\pi}{2}} f(x) =$$

Substitute x with $\pi/2$ in the correct function, then simplify.

$$\lim_{x \rightarrow 0^-} f(x) =$$

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